

Tier 5: Subgroup Fairness Analysis

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0.1 Overview

This report assesses subgroup discrimination and calibration of the Tier-1 nomogram's out-of-sample risk, stratified by sex and age band. External validation on eICU is deferred because no eICU database is present locally; the analysis below delivers the within-MIMIC fairness check that a clinical model needs before deployment. For each subgroup we compute the sample size, event rate, AUROC, and calibration-in-the-large (mean predicted risk versus observed rate).

0.2 Load Libraries

We load the pROC package for computing AUROC within each subgroup.

```
suppressPackageStartupMessages({ library(pROC) })
```

0.3 Locate and Read Data

The helper function searches several candidate paths for the required CSV files. We read the out-of-sample predictions and the landmark dataset (for age and gender), then merge them and create an age-band variable.

```
find_data <- function(f) {  
  cand1 <- c(file.path("../processed_data", f),  
             file.path("processed_data", f), f)  
  hit <- cand1[file.exists(cand1)][1]; if (is.na(hit)) stop("cannot find ", f); hit  
}  
OUT <- dirname(find_data("tier1_oos_pred.csv"))  
  
pred <- read.csv(find_data("tier1_oos_pred.csv"))  
lm6 <- read.csv(find_data("landmark_h6.csv"))[, c("stay_id", "age", "gender")]  
d <- merge(pred, lm6, by="stay_id")
```

```
d$age_band <- cut(d$age, c(-Inf,50,65,80,Inf),
                 labels=c("<50","50-64","65-79","80+"))
```

0.4 Define Subgroup Metrics Function

This function computes, for a given subset of the data, the AUROC, mean predicted risk, observed event rate, and their ratio (calibration-in-the-large). A ratio of 1.0 means the average prediction matches the true event rate; values above 1 indicate over-prediction, and values below 1 indicate under-prediction.

```
grp_metrics <- function(sub, label) {
  y <- sub$onset_within_h; p <- sub$risk_en
  au <- tryCatch(as.numeric(auc(roc(y, p, quiet=TRUE, direction="<"))), error=function(e) NA)
  # calib_ratio: mean predicted risk / observed event rate (calibration-in-the-
  # large) -- 1.0 means the subgroup's average risk matches its true event rate;
  # >1 over-predicts, <1 under-predicts, independent of discrimination (AUROC).
  data.frame(subgroup=label, n=nrow(sub), event_rate=round(mean(y),4),
             auroc=round(au,4), mean_pred=round(mean(p),4),
             obs_rate=round(mean(y),4),
             calib_ratio=round(mean(p)/mean(y),3))
}
```

0.5 Compute Subgroup Metrics and Write Results

We compute fairness metrics for the full cohort, then separately for each sex and each age band. The results are printed and saved to `tier5_subgroups.csv`.

```
rows <- list(grp_metrics(d, "ALL"))
for (g in sort(unique(d$gender)))
  rows[[length(rows)+1]] <- grp_metrics(d[d$gender==g,], paste0("sex=",g))
for (b in levels(d$age_band))
  rows[[length(rows)+1]] <- grp_metrics(d[d$age_band==b,], paste0("age=",b))
tab <- do.call(rbind, rows)
print(tab, row.names=FALSE)
```

##	subgroup	n	event_rate	auroc	mean_pred	obs_rate	calib_ratio
##	ALL	48829	0.0682	0.7752	0.0682	0.0682	1.000
##	sex=F	21519	0.0629	0.7615	0.0631	0.0629	1.003
##	sex=M	27310	0.0724	0.7838	0.0722	0.0724	0.998
##	age=<50	9680	0.0584	0.7561	0.0616	0.0584	1.056
##	age=50-64	14083	0.0670	0.7845	0.0634	0.0670	0.945
##	age=65-79	15960	0.0690	0.7771	0.0680	0.0690	0.985
##	age=80+	9106	0.0790	0.7691	0.0829	0.0790	1.050

```
write.csv(tab, file.path(OUT, "tier5_subgroups.csv"), row.names=FALSE)
cat("\nTier 5 (fairness) complete. External eICU validation deferred (no data).\n")
```

```
##
## Tier 5 (fairness) complete. External eICU validation deferred (no data).
```